REAL TIME MARATHI SIGN LANGUAGE RECOGNIZER USING ML

Major Project Report

Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Engineering (Computer Engineering)

by:

ROHINI RAI	**TU3F2021141**
ARYA PATIL	**TU3F2021150**
AVANTIKA GHARAT	**TU3F2021026**
MAHII VARIAR	**TU3F2021156**

Under the Guidance of

Prof. Pramila Mate

Department of Computer Engineering

TERNA ENGINEERING COLLEGE

Nerul (W), Navi Mumbai 400706

(University of Mumbai)

(2023-2024)

TERNA ENGINEERING COLLEGE, NERUL,

NAVI MUMBAI

Department of Computer Engineering

Academic Year 2023-24

CERTIFICATE

This is to certify that the major project I entitles **"Real Time Marathi Sign Language Recognizer"** is a bonafide work of

Rohini Rai	**ID No: TU3F2021141**
Arya Patil	**ID No: TU3F2021150**
Avantika Gharat	**ID No: TU3F2021026**
Mahii Variar	**ID No: TU3F2021156**

submitted to the University of Mumbai in partial fulfillment of the requirement for the award of the Bachelor of Engineering (Computer Engineering).

Guide **Head of Department** **Principal**

Approval Sheet

Project Report Approval

This Major Project Report – an entitled "**Real Time Marathi Sign Language Recognizer**" by following students is approved for the degree of ***B.E. in "Computer Engineering"***.

Submitted by:

ROHINI RAI	TU3F2021141
ARYA PATIL	TU3F2021150
AVANTIKA GHARAT	TU3F2021026
MAHII VARIAR	TU3F2021156

Examiners Name & Signature:

1._____-_____

2._____-_____

Date:_____

Place: _____

Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Rohini Rai	TU3F2021141	----------
Arya Patil	TU3F2021150	----------
Avantika Gharat	TU3F2021026	----------
Mahii Variar	TU3F2021156	----------

Date: _____

Place: _____

Acknowledgement

We would like to express our sincere gratitude towards our guide ***Prof. Pramila Mate***, Project Coordinators ***Prof. Pramila Mate*** for their help, guidance and encouragement, they provided during the project development. This work would have not been possible without their valuable time, patience and motivation. We thank them for making my stint thoroughly pleasant and enriching. It was great learning and an honor being their student.

We are deeply thankful to **Dr. Kishor Sakure (H.O.D Computer Department)** and entire team in the Computer Department. They supported us with scientific guidance, advice and encouragement, they were always helpful and enthusiastic and this inspired us in our work.

We take the privilege to express our sincere thanks to **Dr. L. K. Ragha** our Principal for providing the encouragement and much support throughout our work.

Rohini Rai	**TU3F2021141**
Arya Patil	**TU3F2021150**
Avantika Gharat	**TU3F2021026**
Mahii Variar	**TU3F2021156**

Date: _____

Place: _____

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Abstract

This project focuses on development of a Marathi Sign Language Recognizer, aimed at overcoming communication barriers encountered by the Marathi-speaking hearing-impaired community in Maharashtra, India. Through the meticulous collection and processing of a diverse array of sign gestures using the MediaPipe framework, a comprehensive and representative dataset has been curated.

Statistical analyses and the strategic application of machine learning methodologies serve as the cornerstone of this endeavor, intending to facilitate precise recognition of Marathi sign language. The ultimate objective is to provide an advanced and efficient system for interpreting these gestures, thereby significantly enhancing communication accessibility for the hearing-impaired. This project, thus, stands as a crucial step in empowering the hearing-impaired community with technology tailored to their linguistic needs.

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List of Abbreviations

Abbreviation	Fullform
CNN	Convolutional Neural Networks
DFD	Data Flow Diagram
GUI	Graphical User Interface
HMM	Hidden Markov Model
IDE	Integrated Development Environment
ISL	Indian Sign Language
k-NN	K-Nearest Neighbours
ML	Machine Learning
MSLR	Marathi Sign Language Recognition
RMMM	Risk Mitigation Monitoring Management
SDLC	Software Development Life Cycle
SLR	Sign Language Recognition
SRS	Software/System Requirement Specifications
UI	User Interface

Chapter 1

Introduction

1.1 Introduction

Let's say you would like to speak with a deaf individual. This may already seem like a challenging task, especially because they are not familiar with sign language. Millions of deaf people struggle with this issue since it prevents them from having conversations with hearing people. The ability to communicate is crucial to human existence. It is difficult to detect the expressions without communication is the main task. This project communicates conservation by recognizing various signals and gestures using webcam and converting them into meaningful message. This recognizer primarily recognizes Marathi Sign Language, which includes the Marathi alphabets, and translates additional words into text or voice. The system handles two-hand gestures, allowing the user to operate it openly as well as enhancing accuracy using the trained dataset. Sign Language Recognition is to detect signs automatically from hand expressions and recognized them accordingly. This sign language Recognition involves converting the signs such as hand gestures and interpreting them to assist close the communication gap with the hearing-impaired persons. This system will assess the signals, actions, and motions and categorize them into messages using a machine learning technique.

A technique that recognises Marathi Sign Language and convert to text will be more helpful for these individuals so they can communicate more effectively with outsiders. The model is trained using convolutional neural networks (CNNs) on the geographical characteristics of the dataset. In MSLRs, words and sentences are formed by static hand morphologies called postures, hand movements called gestures, that correlate to words and sentences. The principal features of this system consist in its ability to identify two-handed gestures and convert them into either text or audio. This Marathi Sign Language Recognizer employs both hand gestures in place of existing ISL, which focuses solely on one hand movement, as a way to improve the number of user interfaces. As a result, it is necessary to examine the specifics of sign descriptions in order to accurately comprehend them and make judgments about the key details of the pictures.

1.2 Organization Of The Report

The present report is methodically structured into six primary sections, which serve to offer an in-depth knowledge of the topic content. The “Introduction” chapter section, which discusses the brief introduction and system's organization, is presented in the First Main section. The “Literature Survey” summarizes the body of knowledge and research in the field, which aids in placing the study in context. It illustrates to how present research fits into the larger academic or practical nature and shows that the project is built upon a foundation of previous work. It focuses on the research papers, problem statement, objectives, scope of the project. Chapter 3 - “Software Analysis and Design” includes the designing content of project. Software model, Proposed System, Hardware and software requirements, the detailed designing which includes all tabular content the diagrams. The RMMM plan is further listed in proper content. The chapter-4 focus the Experimental Setup description of data, along with the data collection methods and methodology used to create Dataset. Chapter 5 includes the “Result and Discussion” which gives us the depth implementation process summarize and analyse research's project contents. In "Conclusion ", the final section of work provides a brief overview of project study, outlining the immense significance of the research findings the ongoing significance of the investigation project."

Chapter 2

Literature Survey

2.1 Existing System Survey

Sign language recognition (SLR) is a computer vision task that involves recognizing hand gestures and facial expressions used in sign language. SLR systems can be used to translate sign language into text or speech. SLRs provide differently abled people a chance to communicate, a chance to get heard. In paper [1] authors Amrutha K, Prabhu P. studied the pre-processing, segmentation, feature extraction, classification and recognition stages systematically. After comparison of various ML based methods, they selected Convex Hull method for feature selection. But the accuracy yielded through this experiment was just 65%. Less accuracy was obtained when the gestures were performed continuously at a fast pace and the small size of the dataset used. The paper [2] mainly emphasizes on the use of Marathi language for dataset creation. The use of deep learning was made to translate the Marathi sign languages to Marathi alphabets (*swar* and *vyanjan*). Also, the system provided around 95% accuracy using right-hand and left-hand gestures simultaneously. But the system's drawback is that the user has to compulsory wear non-skin colored and dark colored clothes otherwise the model's interpretation fails.

The machine translated sign language needs to be given a voice for better understanding. Speech recognition was implemented using python's Sphinx module in paper [3]. This module is used to map the spoken alphabet to text with high accuracy. This paper also offers dynamic gesture detection without loss of accuracy. Although there are a number of advantages this paper offers, improvement is needed when there is a lot of cutter and poor lighting in the environment. In the paper [4] the authors used HMM chain for each gesture and a k-NN model to classify each hand pose. The result was recognition of hand pose in about 0.2s and that for gesture was 0.0037s in real time which was very efficient. Although the paper should include a large and accurate sign language dataset.

One of the best ways to communicate freely is without pretending and without changing ourselves as per the systems environment. The system should be able to identify our hand gestures in the natural way. This same concept was implemented successfully in paper [5] where bare hand and facial features were decoded smoothly. Offline sign detection is also

possible with this system. Despite all this the paper only considers one hand sign detection. Concept of considering facial features, emotions while detecting a sign was stated in paper [6] by authors Razieh Rastgoo, Kourosh Kiani, Sergio Escalera. The system stated in the paper was never implemented thus the success rate of this system is not known.

2.1 Problem Statement

- SLR systems can be sensitive to variations in the environment, such as changes in lighting or the presence of background noise. This impacts the systems' performance in diverse environments leading to poor accuracy.
- SLR systems are computationally expensive to train and deploy. The demand for such systems is rising but the development rate is very low.
- Sign detection systems especially for Marathi language are very few.
- Existing SLR systems often struggle with recognizing sign language dialects and regional variations. There is a need to overcome these variations and be more adaptable.
- Different types of users will operate the system. These users may have unique signing styles and preferences which can lead to individual differences.

2.2 Objectives

- To identify symbolic gestures used in communication and convert it into text.
- To develop a model that can handle a large amount of sign language expression data using Python.
- To provide simultaneous translation and detection for facilitating smooth interaction and communication along with corresponding labels.
- To put effort into accurately recognizing as well as interpreting and translating the visuals into meaningful expressions.

2.3 Scope

- MSLR has a large scope in:

i. Education: In the education field, sign language recognition technology aids students who are deaf or hard of hearing by providing real-time sign language interpretation of lectures and educational content, ensuring their equal access to quality education.

ii. Employment: Sign language recognition enhances workplace accessibility by allowing

deaf employees to effectively communicate with colleagues, superiors, and customers, fostering inclusivity and equal job opportunities.

iii. Communication: It enables real-time sign language interpretation for video calls and public events, breaking down language barriers and ensuring that deaf individuals can fully participate in various communication contexts.

iv. Entertainment: Sign language recognition can be utilized in the entertainment industry for automatic sign language interpretation during live broadcasts or events, making entertainment content more accessible to the deaf and hard of hearing audience.

Chapter 3

Software Analysis and Design

3.1 Software Model

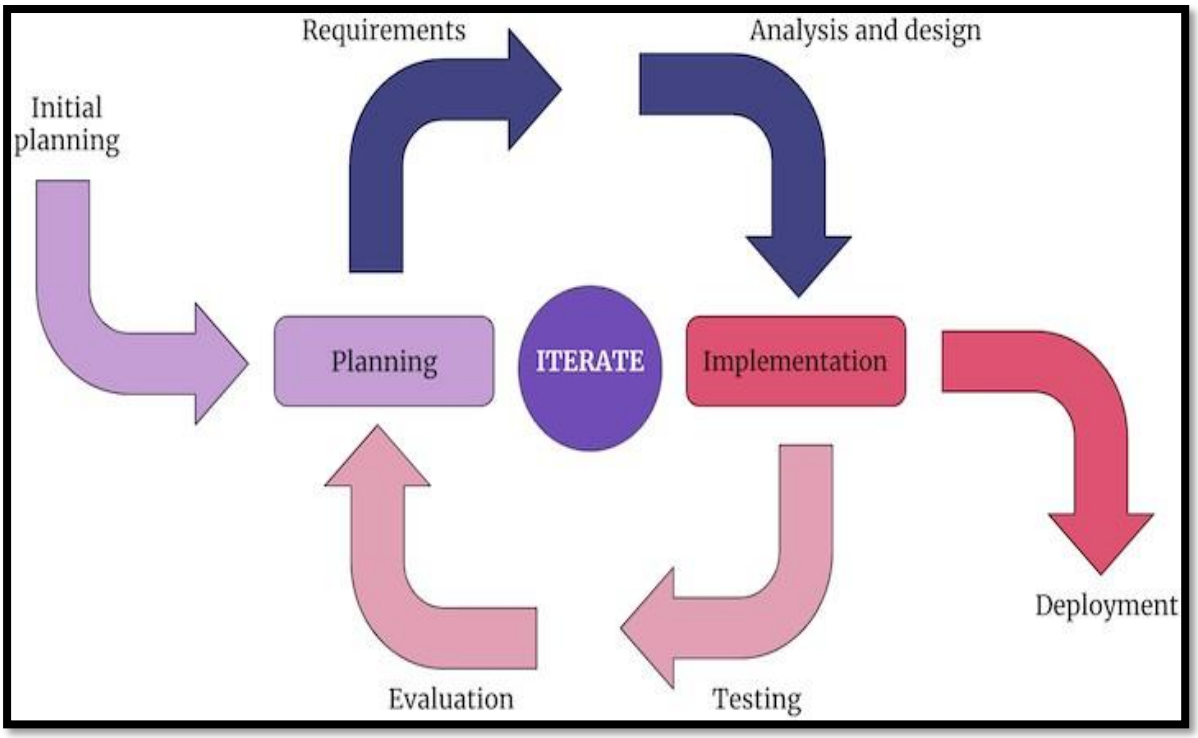


Fig. 3.1 Iterative and Incremental Model

About the Iterative and Incremental Model SDLC Model:

- Description: This model involves breaking the project into small parts and developing them in iterations, each adding new features or refining existing ones. It allows for flexibility and adjustments as the project progresses.
- The Iterative and Incremental Model is an agile software development approach that breaks down a project into small, manageable parts called iterations. Each iteration goes through the entire development cycle, from planning to deployment, with the goal of refining and adding features.
- For the Marathi Sign Language Recognition project, the Iterative and Incremental Model is ideal as it enables regular user feedback, progressive refinement of the recognition model, and parallel development of different project aspects. It ensures

adaptability to evolving user needs and facilitates a faster time-to-market for essential features.

3.1.1 Phases of Software Model

Requirements:

- This phase is about gathering and analyzing the requirements for the system. This includes identifying the needs of users and stakeholders, as well as the functional and non-functional requirements of the system.

Initial planning:

- This phase is about creating a high-level plan for the project. This includes identifying the number of iterations, the features that will be delivered in each iteration, and the resources that will be needed.

Planning:

- This phase is about developing a detailed plan for the iteration. This includes identifying the specific tasks that need to be completed, the resources that will be needed for each task, and the estimated time to complete each task.

Analysis and design:

- This phase is about developing a working architecture for the system and designing the system components. This includes developing use cases, activity diagrams, and sequence diagrams.

Implementation:

- This phase is about coding the system components and integrating them into a working system. This includes developing unit tests and integration tests.

Evaluation:

- This phase is about testing the system to ensure that it meets the requirements and that it is ready for release. This includes conducting system testing and acceptance testing.

Testing:

- This phase is about deploying the system to a test environment and conducting performance testing and user acceptance testing.

Deployment:

- This phase is about deploying the system to production and making it available to users.

Feedback:

- This phase is about collecting feedback from users and using it to improve the system in subsequent iterations.

The iterative and incremental model is a cyclical process, with each phase being repeated in each iteration. Feedback from users and stakeholders is used to improve the system in each iteration. This ensures that the system meets the needs of users and that it is delivered in a timely and efficient manner.

3.2 Proposed System

Creating a system for Marathi Sign Language that also converts text-to-speech involves several steps. First, we need to collect a diverse dataset of Marathi Sign Language signs and gestures, ensuring it covers a wide range of communication scenarios. This data must be cleaned and preprocessed to remove any noise or inconsistencies. Next, we extract important features from the signs, like hand shapes, movements, and gestures. These features are used to train deep learning models, like Convolutional Neural Networks enabling the system to recognize signs accurately. Then, we teach computers to understand these signs and translate them into words. We also make sure the computer can say these words out loud. People can use a simple interface to communicate using sign language, and the computer will communicate on their behalf. We test it with sign language users and keep making it better. It should work for everyone and keep information safe. If more people use it or technology changes, it can handle that. We also need to keep it updated and teach people how to use it.
Regular maintenance and updates keep the system current, while outreach and education efforts ensure it benefits the Marathi Sign Language community and the public at large.

3.3 System Requirement Specification (SRS)

3.3.1 Intended Audience

This recognizer is intended for end users, developers, data analyzer, researchers, project managers, data providers, data collectors, testers. This system is mainly intended to the dumb/deaf or hearing-impaired individuals and wider community to communicate via using Marathi Language.

3.3.2 Product perspective

The proposed software will be able to help overcome the huge communication barrier between Sign Language users and verbal speakers in Maharashtra. Our Marathi sign language recognition system will help to convert the hand gestures used in Marathi sign language into text as well as speech form. The existing software applications that help in sign language conversion mostly focus on American Sign Language and to some extent Indian Sign Language. Our proposed system focuses on converting Marathi Sign Language gestures to text and speech format.

3.3.3 Functional requirements

- An image sensor captures the gestures made by the sign language user.
- OpenCV tracks the captured gestures by identifying feature points.
- The captured data is preprocessed.
- The preprocessed data is fed to the model for training and testing.
- Based on the data provided, the text will be predicted.
- The text selected will be the one with the highest possibility of up to three words.
- Predicted text will be displayed on the User Interface/screen.
- Once text is predicted it will be converted to speech/voice format.
- Users will have the option to turn on speech format for the predicted texts.

3.3.4 User characteristics

This project will be useful to people who have speech disabilities and are comfortable using the Marathi sign language to communicate. This project is useful for:

- People with speech disability
- People with hearing disability
- Speech impaired people who can communicate using Marathi sign Language.
- People who have basic knowledge of operating a computer system
- Non-speech impaired people who cannot understand sign language.

3.3.5 Design constraints

- The system works well when hand gestures are performed at a slow speed, but accuracy decreases if the hand gestures are too fast.
- Software may not efficiently work on mobile phones which lack the minimum hardware requirement to run this software.
- Host device must have camera/sensor with minimum 720p resolution.
- Hardware limitation: Minimum of 4GB RAM required.
- Good quality microphone.
- With the current UI of the project only one way communication is possible.

3.3.6 User Documentation

The software and GUI of this project will be basic and easy to use for users of any level of digital awareness. Users simply require a basic understanding of using computers to be able to use this software.

3.3.7 Assumptions and Dependencies

Following are the assumptions taken for the system:

- It is assumed that the user will have an embedded or external camera or image sensor available on their host device to be able to capture the hand gestures.
- It is assumed that the user will be running the software on a system that meets the minimum hardware requirements as stated before in Hardware and Software requirements.

Following are the dependencies of the system:

- OpenCV is a dependency of this project.
- Mediapipe is a dependency of this project.
- Python and its NumPy library are dependencies of this project.

3.3.8 User Interface

The user interface of Marathi Sign Language Recognizer is very intuitive as it provides a user-friendly experience for both sign language users and the individuals interacting with them. This software application aims to bridge the gap between dumb/deaf or hearing impaired and other community.

3.3.9 Supplementary Requirements:

When creating a plan for a Marathi Sign Language Recognizer, along with the main requirements, we need to consider a few extra things. We must decide how people will interact with the system, gather a diverse set of Marathi Sign Language gestures, figure out which signs the system needs to understand, make sure it recognizes signs accurately and quickly, adapt to how different people sign, work well in different settings, show result in real-time, work on different devices, keep people's information safe, handle mistakes, be easy for everyone to use, explain how to use it, test it thoroughly, think about different ways people might use it, and be ready to update and improve it over time.

3.3.10 Non-Functional requirement:

Performance requirements:

- The system should be able to accurately recognize the signs and gestures in real time.

Security requirements

- The system should be secure and protect the user's private data.

Portability requirements

- The system can be executed on computer systems having any versions of Windows operating system, macOS.

Maintainability requirements

- The system should be maintained and controlled only by authorized admins.

Reliability requirements

- The system should be able handle variations in lightning, difference in skin tones, background noises and other environmental factors without affecting the recognition accuracy.

Usability requirements

- The user interface should be easy to understand as well as use for both sign language users and non-sign language users.

3.4 Hardware and Software Requirements

3.4.1 Hardware specification:

- Camera with minimum of 720p resolution.
- Microphone for speech recognition
- A modern multi-core processor, such as Intel Core i5 or equivalent.
- A minimum of 4GB RAM.
- A minimum of 500 MB of available storage space is recommended.

3.4.2 Software specification :

- The system should be compatible with mainstream operating systems such as Windows, macOS, Linux or Ubuntu.
- The IDE is Visual Studio Code (VS code)/ PyCharm
- The language requirement is Python.
- The libraries and frameworks requirements are OpenCV, Mediapipe, Scikit, TensorFlow, Keras.
- 64-bit Windows Operating System is used.

3.5 Design

3.5.1 Data Flow Diagrams

DFD Level – 0: Describes the basic operation of the proposed system.

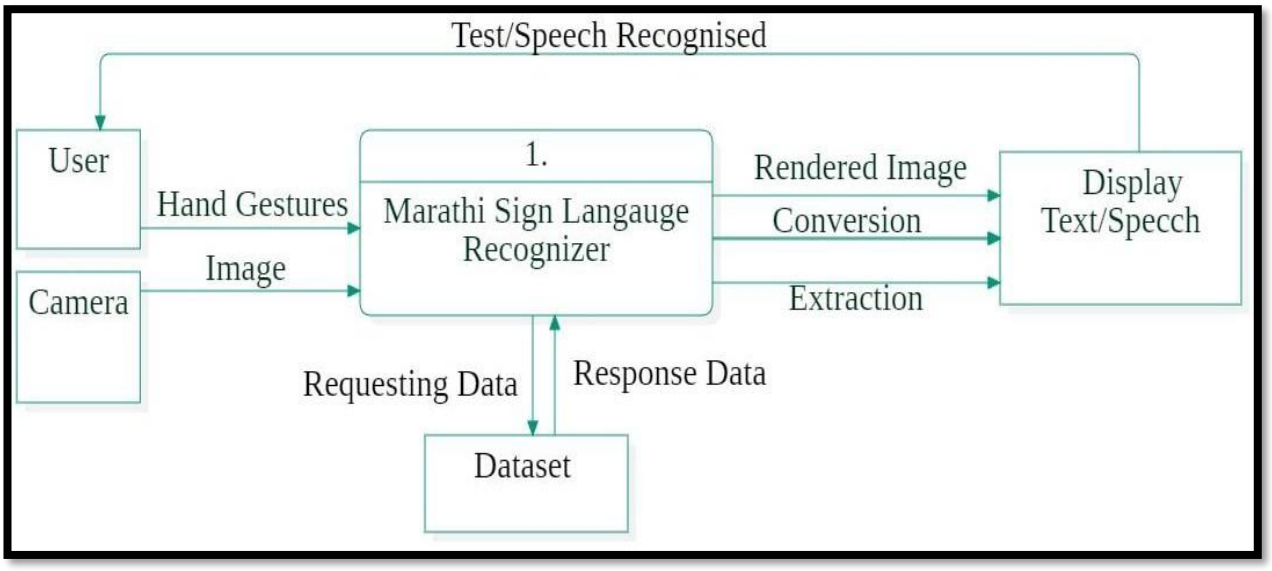


Fig. 3.2 DFD level 0

DFD Level – 1: Describes briefly the operations of the proposed system.

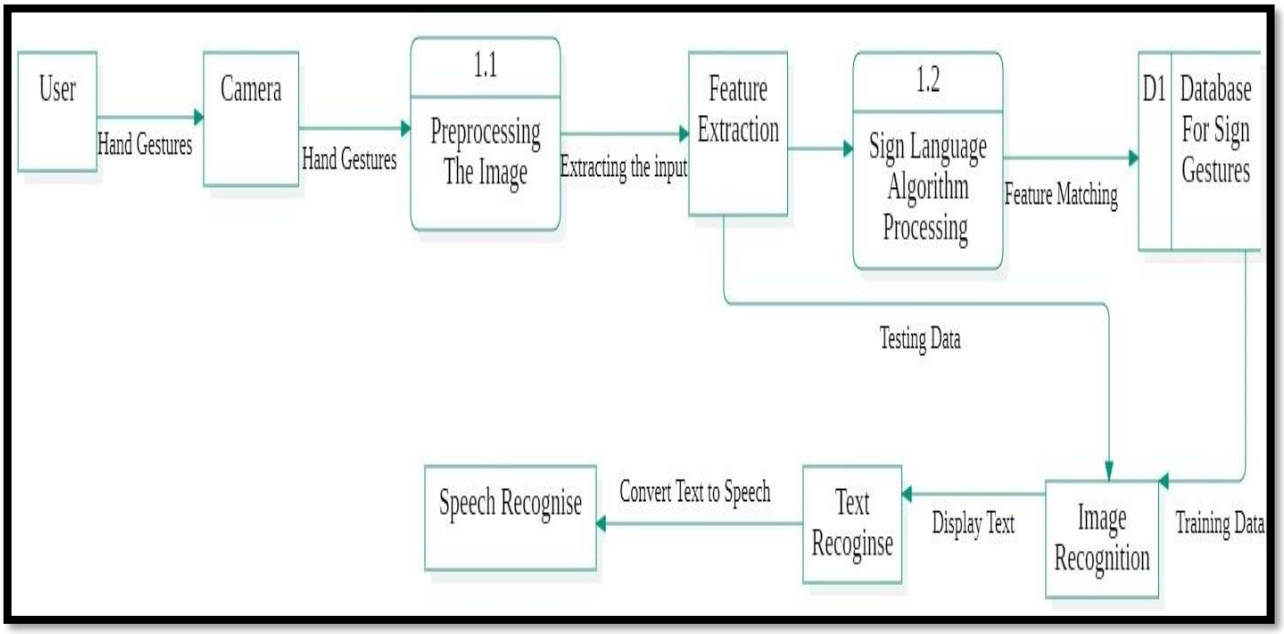


Fig. 3.3 DFD level 1

DFD Level – 2: Describes the detailed operation of the proposed system.

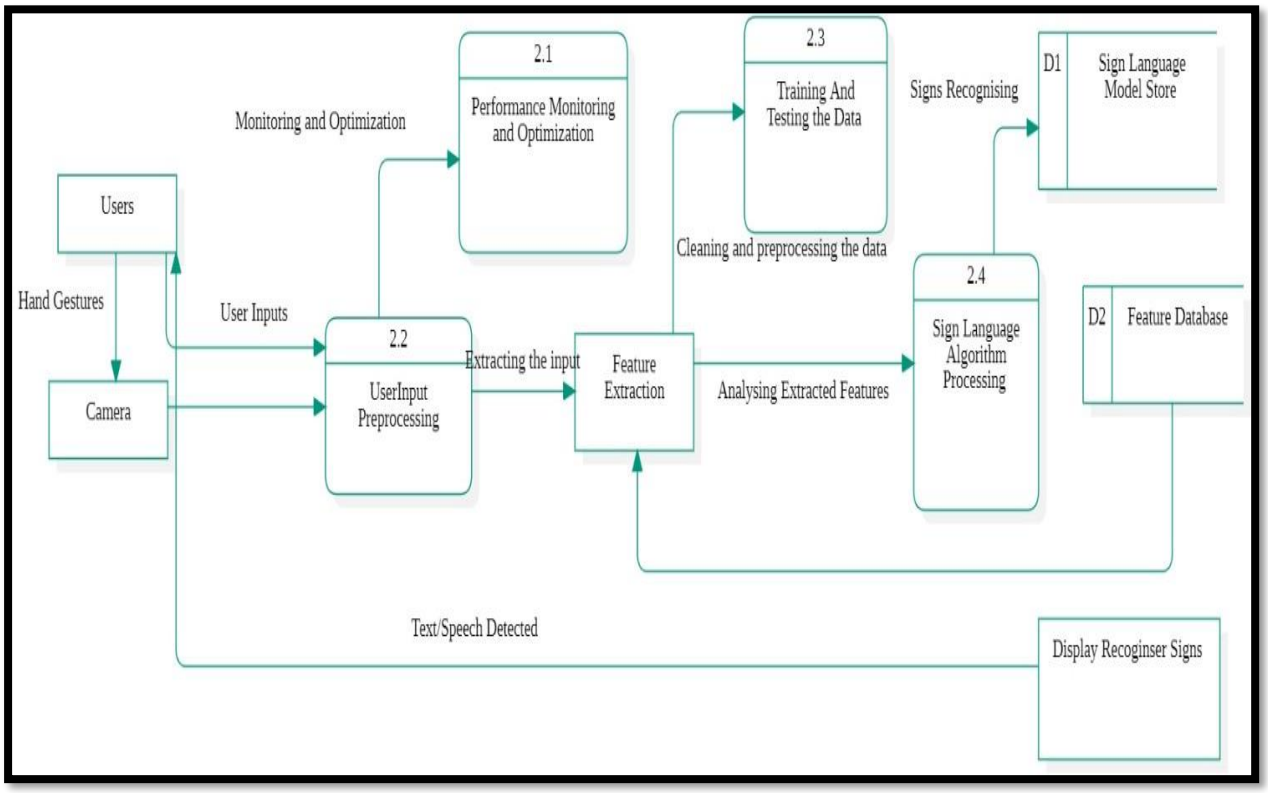


Fig. 3.4 DFD level 2

3.5.2 Use Case Diagram

The following diagram state the usecase of the proposed system:

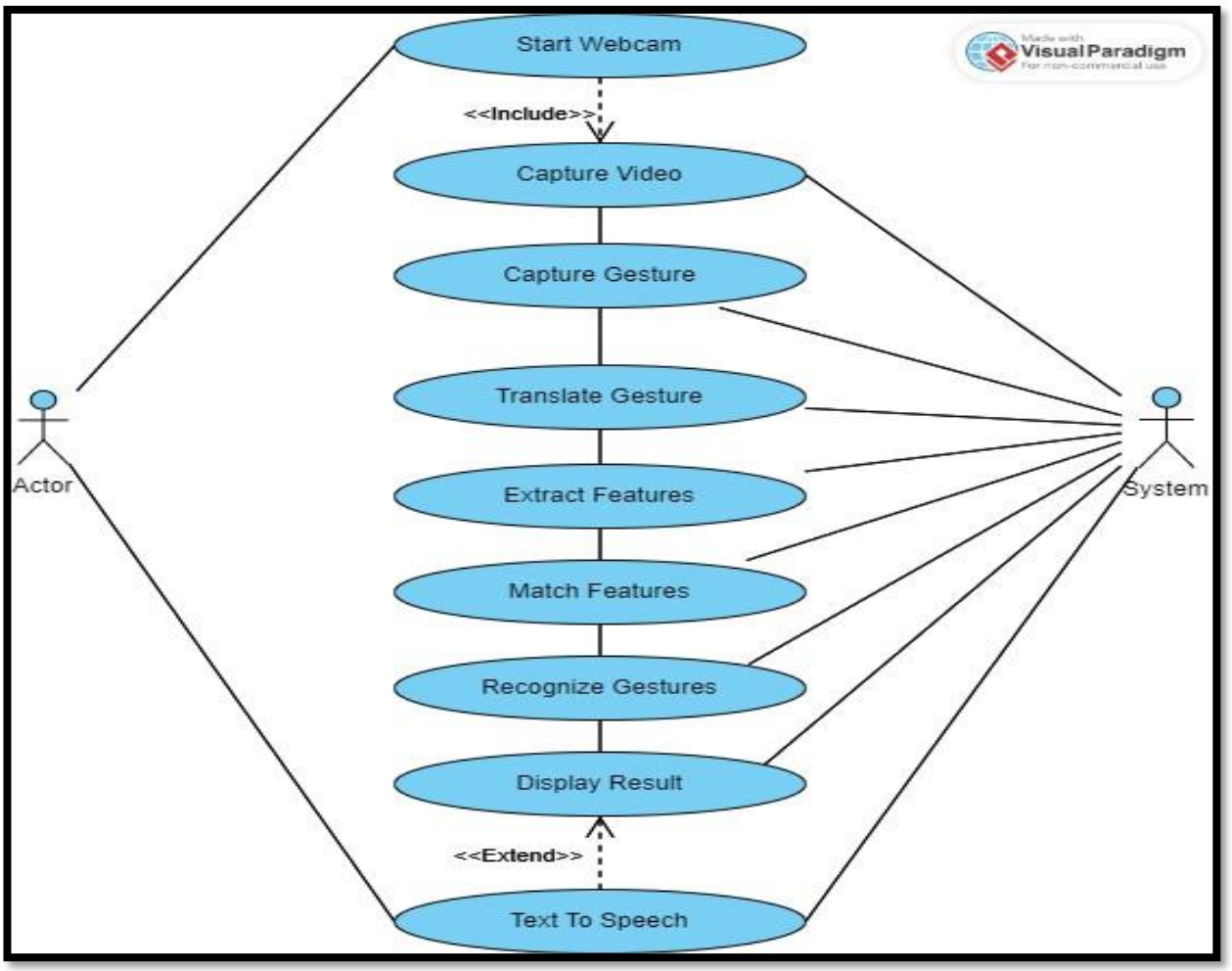


Fig. 3.5 Use case Diagram

3.5.3 Flowchart Diagram

The following diagram shows the flow of the proposed system:

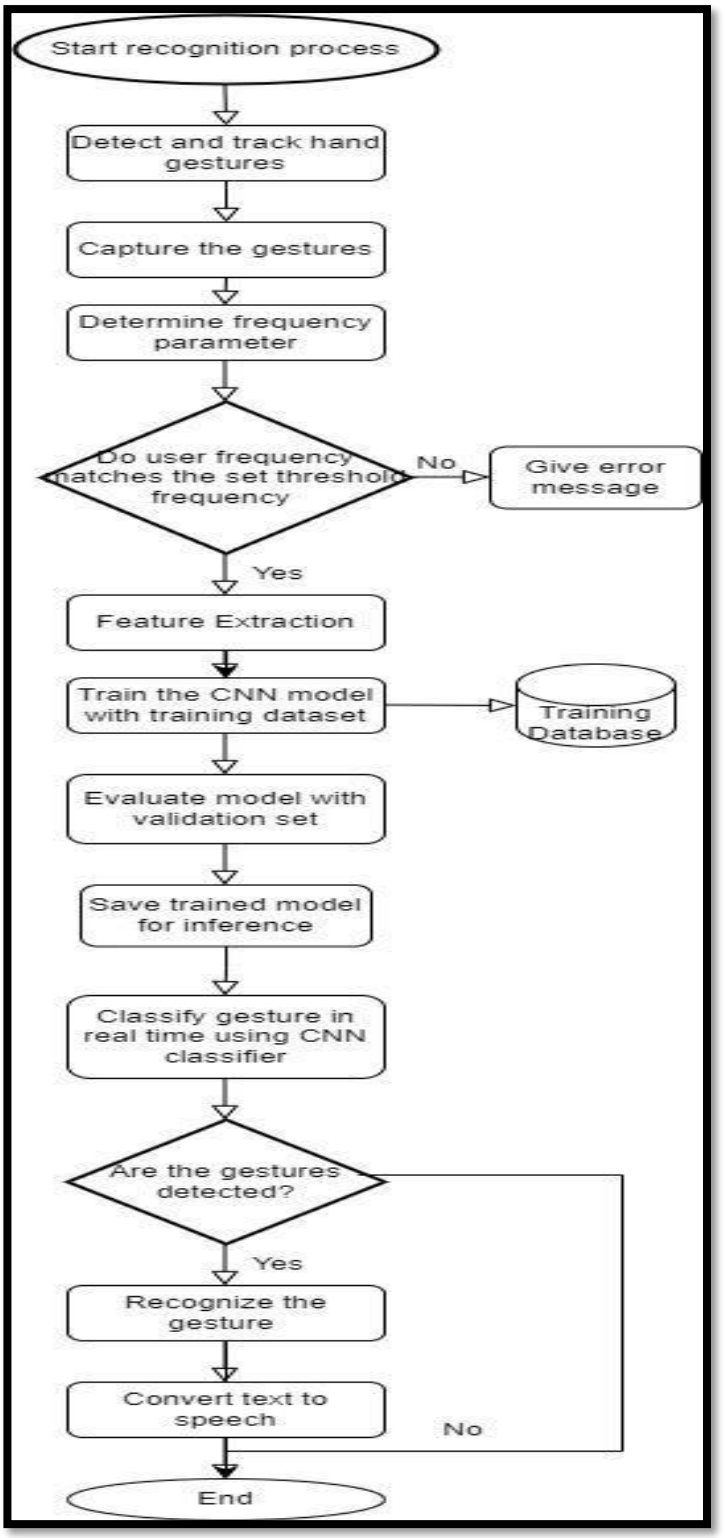


Fig. 3.6 Flowchart Diagram

3.5.4 Sequence Diagram

The following diagram describes the sequential flow of the proposed system:

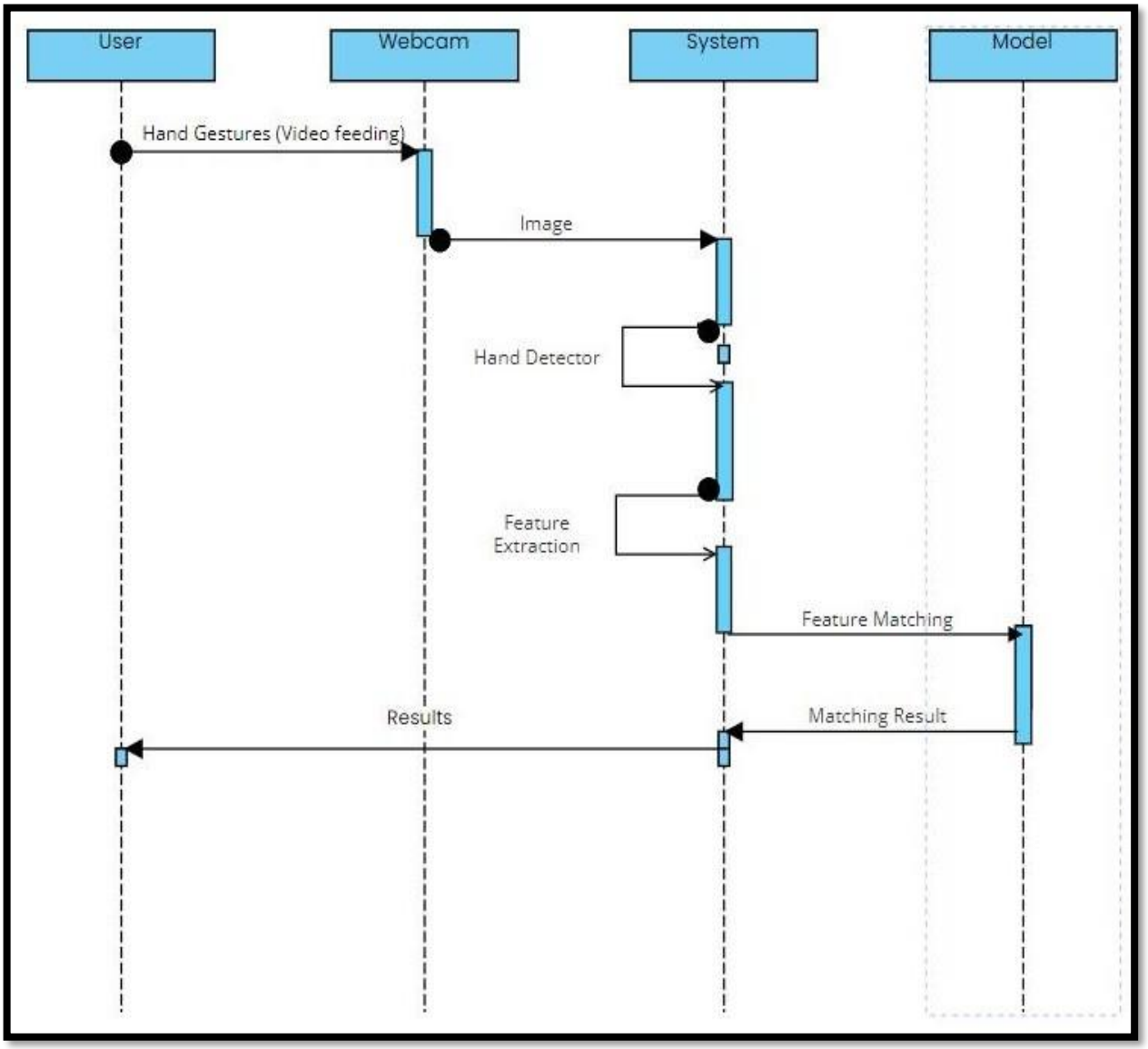


Fig. 3.7 Sequence Diagram

3.5.5 Activity Diagram

The following diagram shows the various activity that takes place in the proposed system:

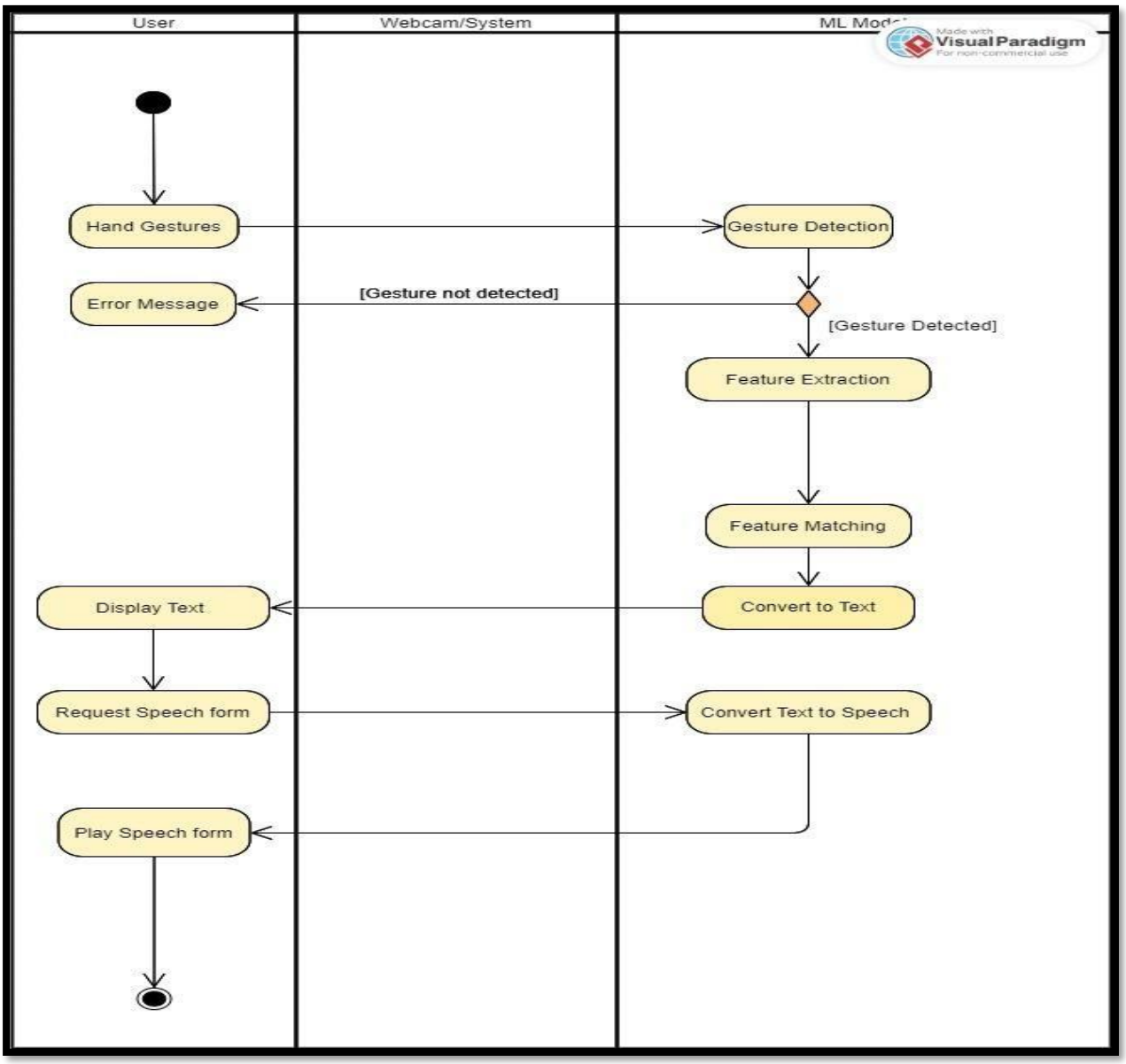


Fig. 3.8 Activity Diagram

3.6 Risk Mitigation Monitoring and Management Plan

1. Risk Identification:

Technical Challenges:

- Description: Unforeseen technical complexities in Marathi Sign Language recognition algorithms like limited availability of labelled data, user interaction design
- Potential Impact**:** Delays in project timelines, increased development costs.

- Probability: High
- Mitigation: Conduct a comprehensive literature review and feasibility study before implementation. Collaborate with experts in computer vision and sign language recognition. Include feedback from the users

Data Quality Issues:

- Description: Inadequate or inconsistent sign language data.
- Potential Impact: Reduced accuracy in recognition, biased model outcomes.
- Probability: Low to Medium
- Mitigation: Verify data quality through preprocessing techniques. Collect diverse and representative samples of Marathi Sign Language gestures.

Budget Constraints:

- Description: Unanticipated costs in hardware, software, or resources or shortage of resources.
- Mitigation: Regularly review the project budget. Seek alternative resources or funding if needed.

2. Risk Assessment:

High/Medium/Low:

- Technical Challenges: High
- Data Quality Issues: Medium
- Budget Constraints: Low

Probability and Impact:

- Technical Challenges: Moderate probability with high impact
- Data Quality Issues: Moderate probability with medium impact
- Budget Constraints: Low probability with medium impact

3. Risk Response:

Technical Challenges:

- Response: Maintain close communication with project mentors and college faculty. Regularly update project timelines to accommodate unexpected challenges.

Data Quality Issues:

- Response: Implement rigorous data validation processes. Establish clear criteria for

data inclusion.

Resource Constraints:

- Response: Monitor project expenses closely. Have alternative resources ready and maintain open communication with the mentors.

4. Risk Monitoring:

- Establish a regular risk review meeting.
- Use project management tools to track progress and identify potential risks.
- Encourage team members to report any emerging risks promptly.

5. Contingency Plan:

- Identify alternative algorithms or technologies in case of major technical challenges.
- Establish backup data sources in case of data quality issues.
- If possible, make additional dataset available in case of
- Identify potential additional funding sources in case of budget constraints.

6. Communication Plan:

- Keep constant communication with group members and also with your project mentors
- Regularly communicate project risks and mitigation strategies to stakeholders.
- Develop a communication protocol for notifying stakeholders in case of significant risk events.

7. Documentation:

- Maintain a comprehensive risk log documenting identified risks, their assessment, and responses.

8. Risk Ownership:

- Clearly define responsible parties for monitoring and addressing specific risks.

Chapter 4

Methodology

4.1 Experimental Setup

- **Methodology Flow :**

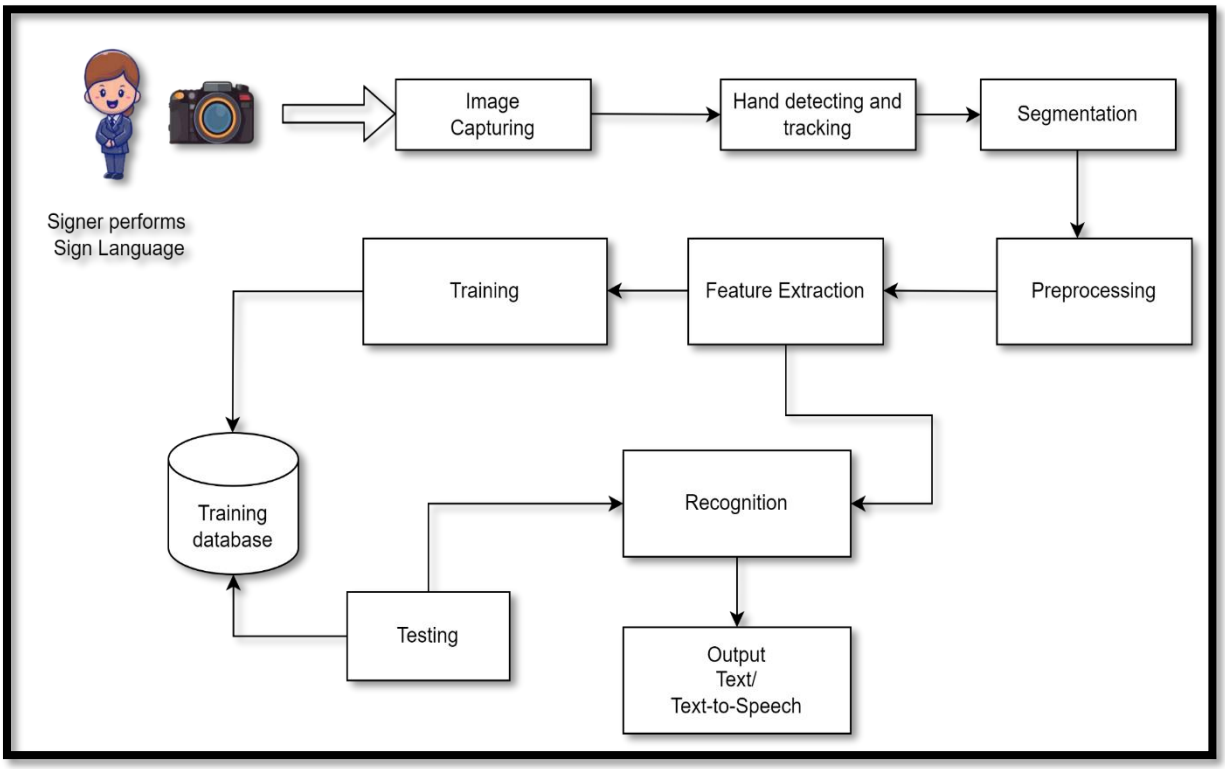


Fig. 4.1 Methodology Flow

4.1.1 Description of Data

The dataset for the Marathi sign language recognizer project was generated by us using the MediaPipe framework. Due to the unavailability of Marathi sign language datasets online, we took the initiative to create our dataset. The collected data is primary, obtained through video recordings of individuals performing various Marathi sign language gestures. These recordings were then processed using MediaPipe to extract key points and form the dataset.

The data collection process involved recording individuals performing a wide range of Marathi sign language gestures. We ensured diversity in gestures to create a comprehensive dataset that covers various movements and expressions used in the Marathi sign language. Given the lack of existing datasets, a purposive sampling technique was employed, aiming to include a broad range of gestures, ensuring representation of commonly used signs in Marathi sign language.

4.1.2 Data Collection Methods

The data collection process involved recording individuals performing a wide range of Marathi sign language gestures. We ensured diversity in gestures to create a comprehensive dataset that covers various movements and expressions used in the Marathi sign language. Given the lack of existing datasets, a purposive sampling technique was employed, aiming to include a broad range of gestures, ensuring representation of commonly used signs in Marathi sign language.

4.1.3 Data Description

Variables: The dataset includes coordinates and temporal data of key points extracted from the MediaPipe framework. These points represent various body parts and movements involved in signing gestures.

Data Types and Format: The data is primarily composed of numerical values representing the spatial coordinates of key points extracted from the videos. The dataset is stored in a structured format, compatible with MediaPipe's output specifications.

Data Size and Characteristics: The dataset consists of 'N' recordings, each containing 'M' frames, where each frame captures the spatial coordinates of 'K' key points. The dataset has been reviewed for any inconsistencies or missing values, ensuring its quality for subsequent analysis.

4.2 Methodology Used to Create Dataset

4.2.1 Data Cleaning

Preprocessing involved the removal of any noisy or irrelevant frames, eliminating possible errors or artifacts generated during the video processing. Additionally, any missing data points were addressed through interpolation techniques to maintain the dataset's integrity.

4.2.2 Transformation and Normalization

The dataset underwent normalization procedures to standardize the coordinates and scale, ensuring consistency across all gesture samples.

4.2.3 Dataset preparation

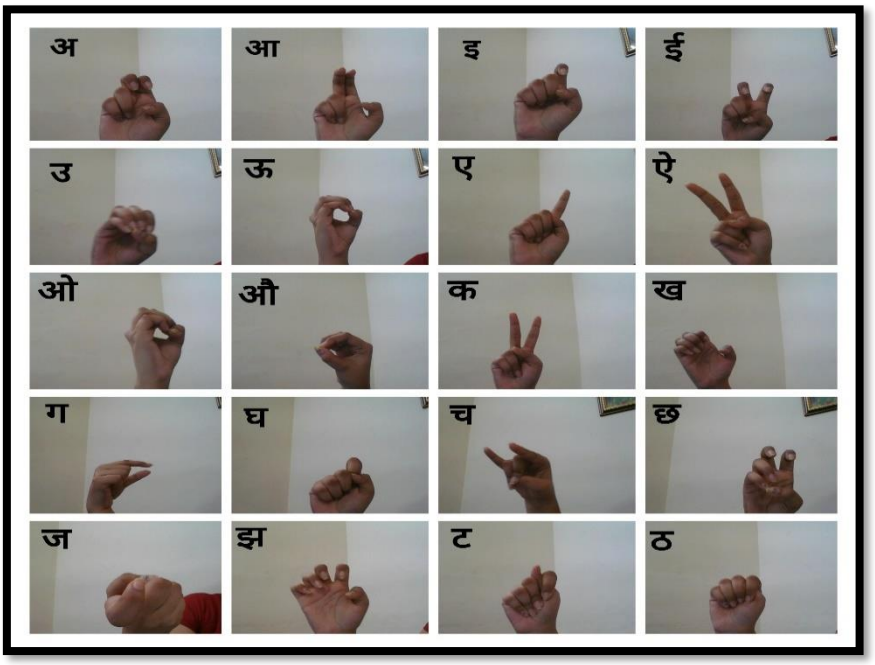


Fig. 4.2 Created Dataset

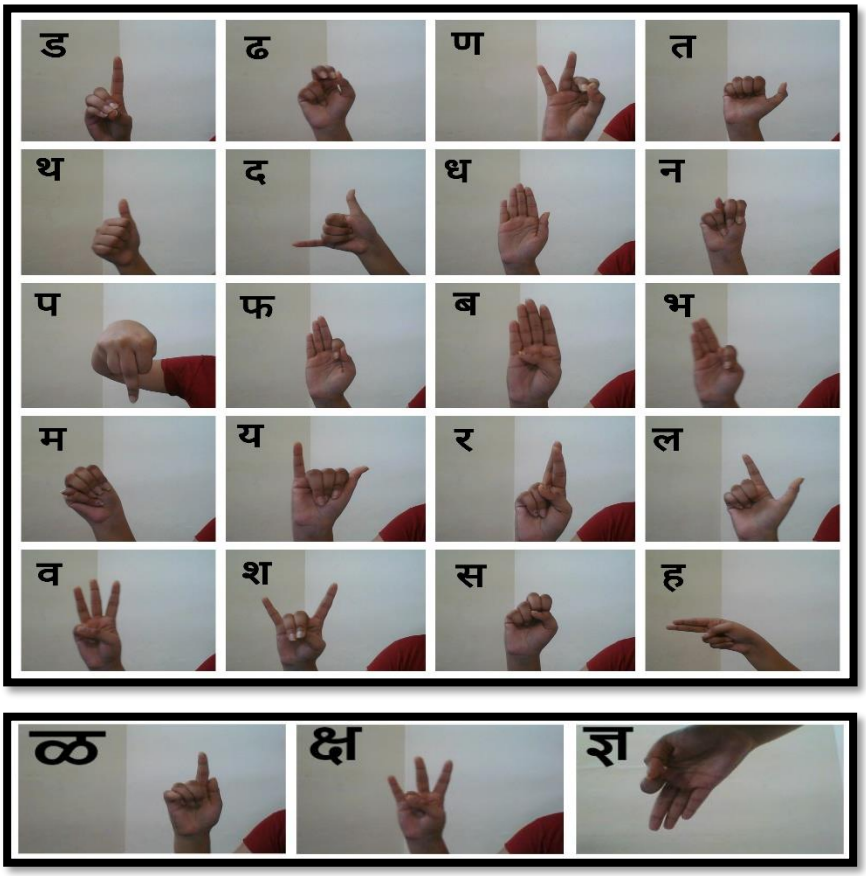


Fig. 4.3 Created Dataset

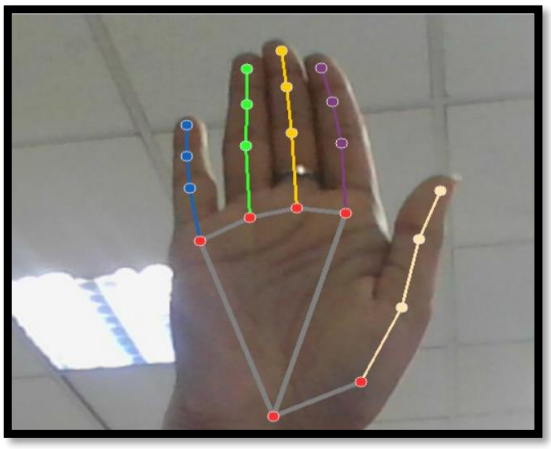


Fig. 4.4 Created Dataset

The 21 landmark points is considered, would be strategically placed on the hand to represent key features of the signs. These landmark points could include positions on the fingers, palm, and other relevant regions of the hand.

Chapter 5

Problem Timeline

5.1 Gantt Chart (timeline chart)

- **SEM 7:**

The following is Timeline of Sem 7 describing the tasks completed in Sem 7:

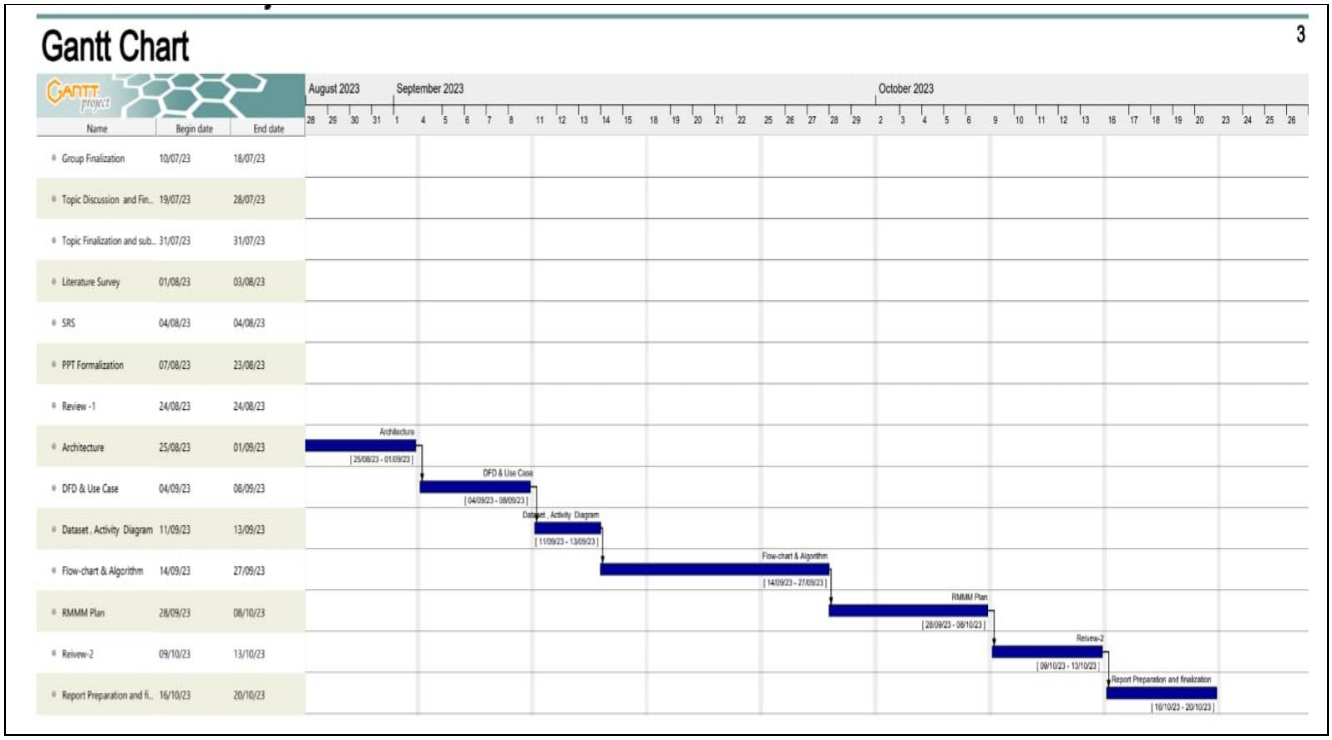


Fig. 5.1 Timeline SEM 7

- **SEM 8:**

  The following is Timeline of Sem 8 describing the tasks to be completed in Sem 8:

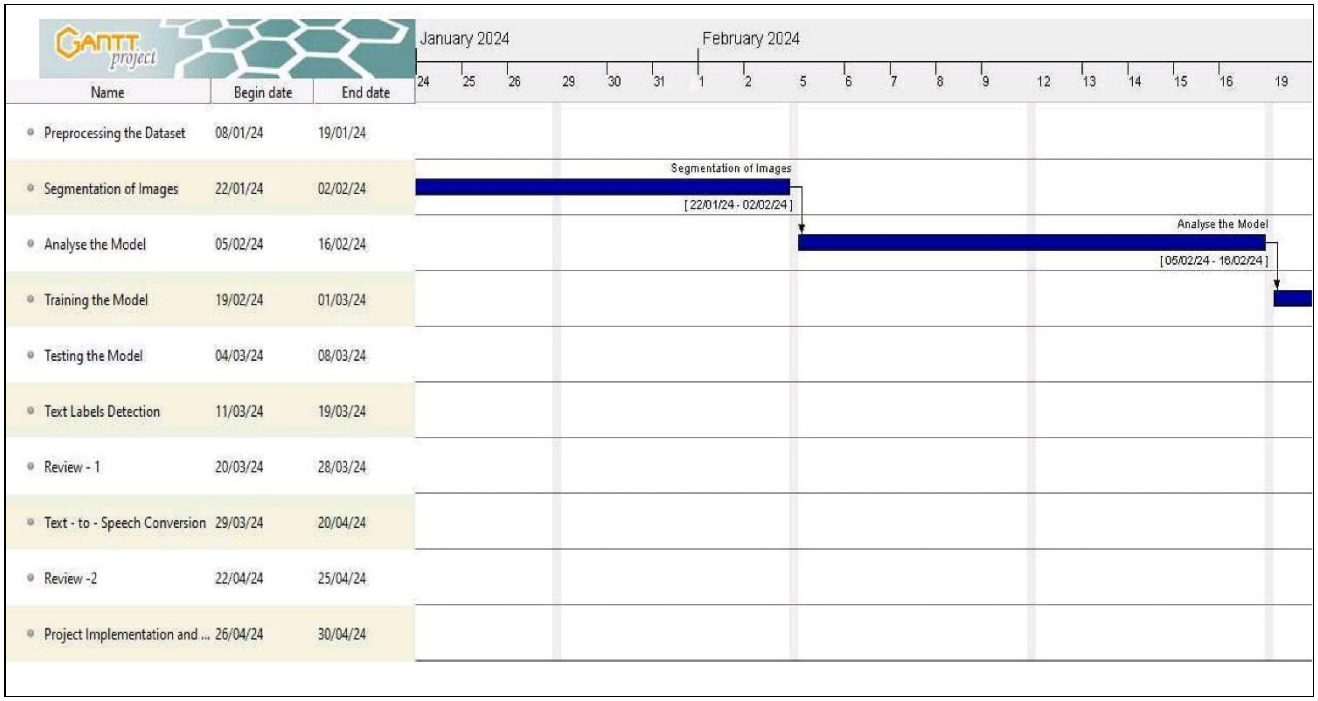


Fig. 5.2 Timeline SEM 8

Chapter 6
Implementation

6.1 Screenshot of the output :

6.1.1 The User Interface that pops up after running the application.

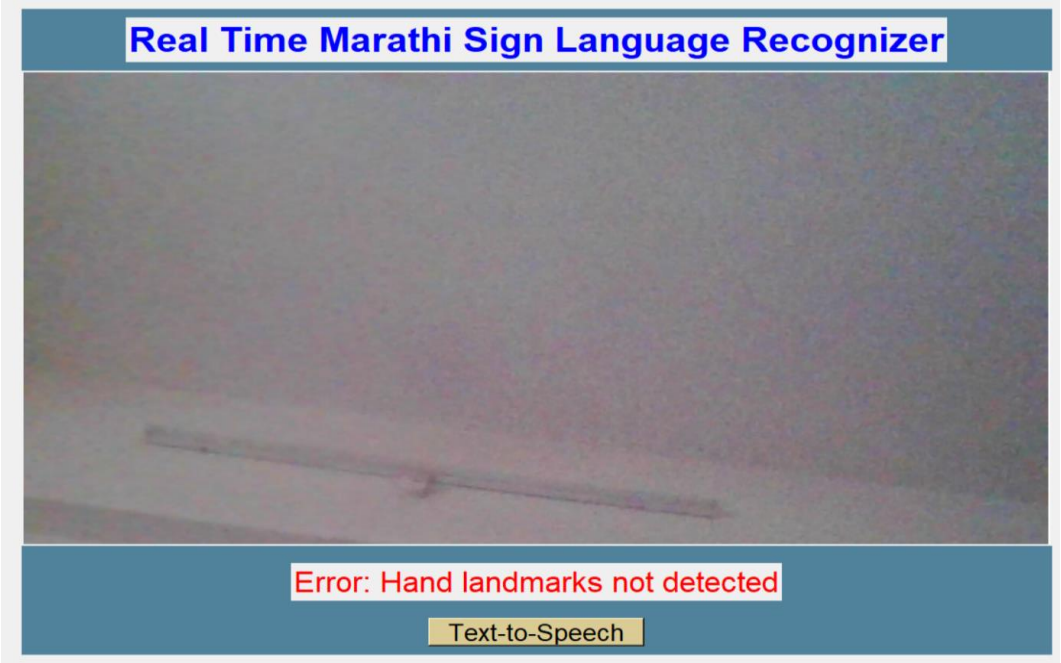


Fig. 6.1 User Interface(1)

6.1.2 If I am showing any gesture which does not exists in the dataset then it will give a message saying “Hand gesture not predicted”.

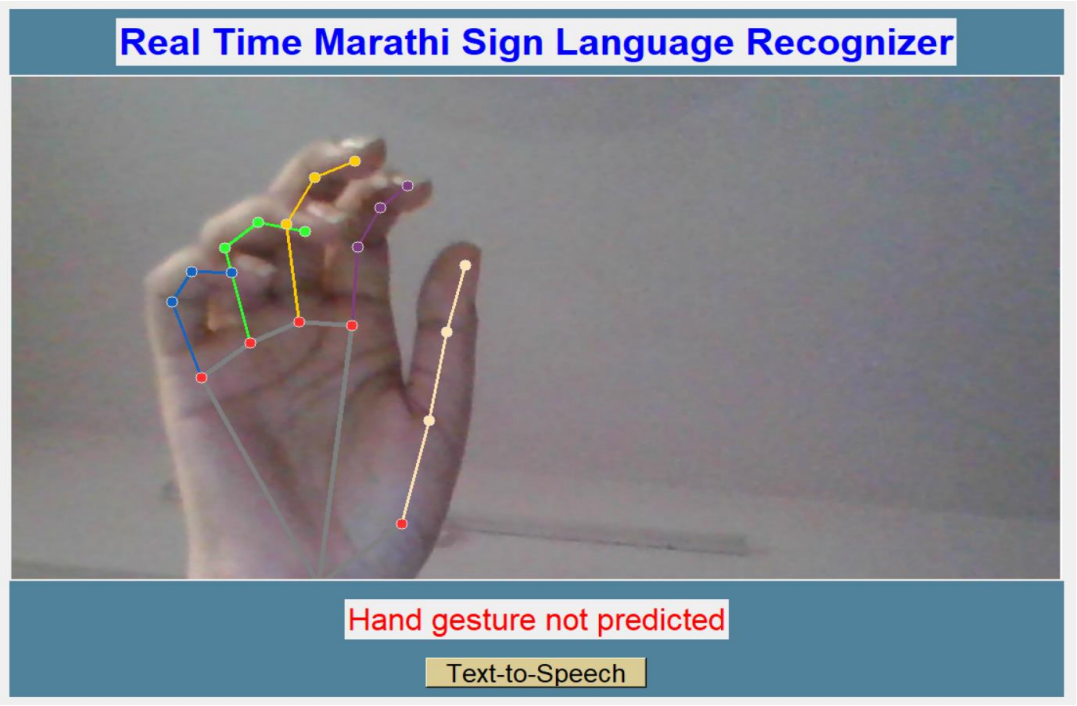


Fig. 6.2 User Interface(2)

6.1.3 If the system correctly predicts the hand gesture which is available in the dataset then it will show a label or a text corresponding to that image from the dataset.

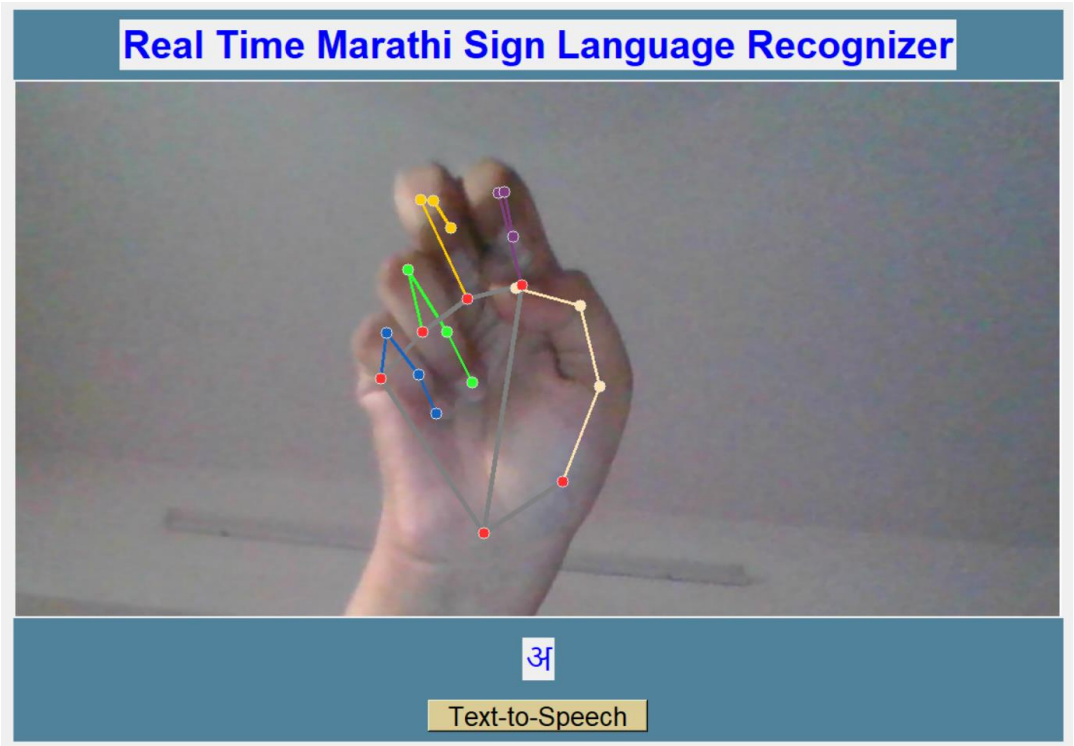


Fig. 6.3 User Interface(3)

6.1.4 Similarly, it can also predict the words or phrases which are available in the dataset.

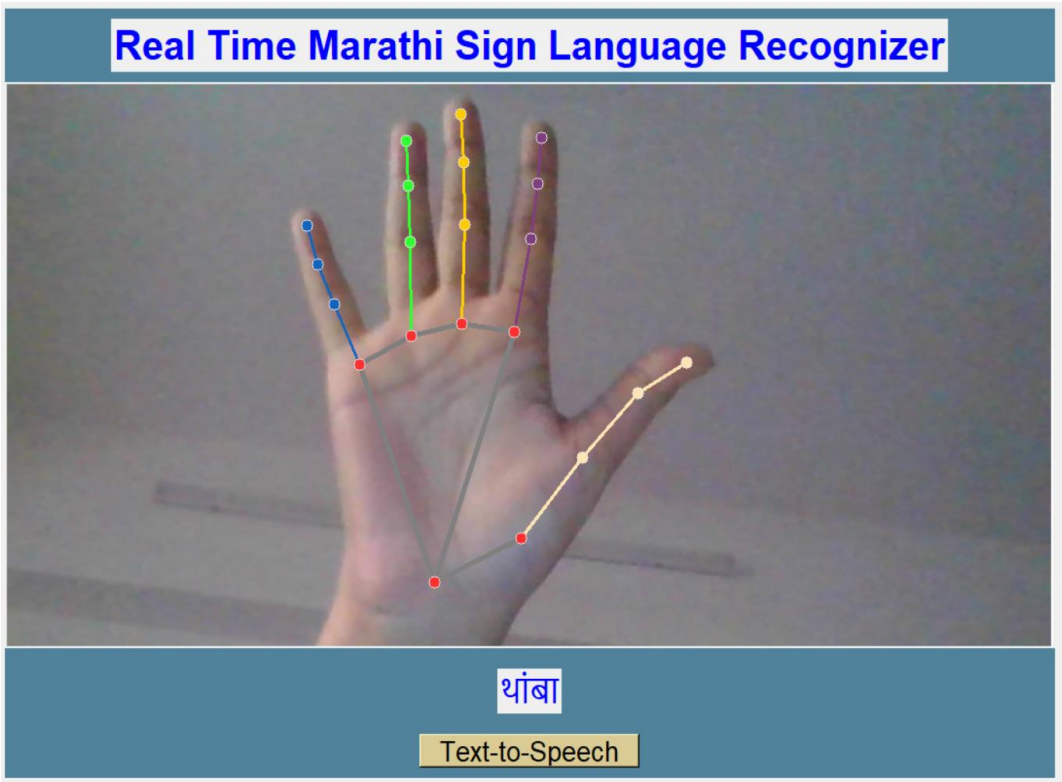


Fig. 6.4 User Interface(4)



Fig. 6.5 User Interface(6)

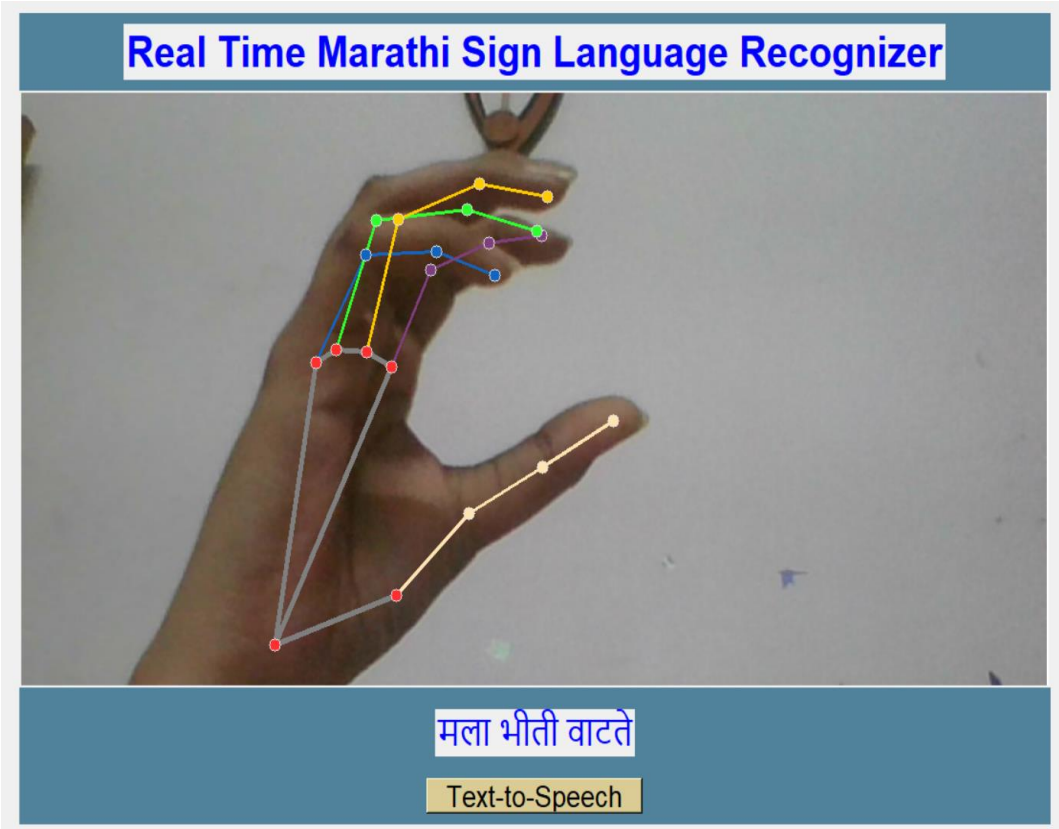


Fig. 6.6 User Interface(7)

The user can also convert the text into speech format by clicking on the “Text-to-Speech” button.

Chapter 7

Result and Discussion

7.1 Result Table :

Algorithm	Accuracy score	Precision score	Error score
CNN (Convolutional Neural Network)	98.38%	0.97	1.16%
SVM (Support Vector Machine)	96.30%	0.95	3.50%
Random Forest Algorithm	94.40%	0.93	4.95%
KNN (K-Nearest Neighbors)	91.56%	0.90	8.45%

Fig. 7.1 Result Table

The above figure (7.1), shows the comparison of the algorithms which best fits the model with respect to their Accuracy score, Precision score and Error Score. From this we concluded that CNN (Convolutional Neural Network) gives the best accuracy which is “98.38%”. Also CNN was a suitable algorithm for our dataset because our dataset contains an images.

7.2 Result and Discussion :

1. Better outcomes can be achieved by having the signer perform precise and correct signsin front of a webcam.
2. In order to facilitate frequent communication between users, the system focuses on twoprimary hand gestures movements.
3. It will be able to focus on turning a sequence of movement into sentences, and then speech and text.
4. During translation resizing of image, extraction, noise reduction, translation into

textual content or audio and then comparison with database is done.

5. Depending on the signer's hand size and technique when making signs in front of the camera, the system produces the best possible outcomes.
6. Discuss the real-world applications and benefits of an accurate Marathi Sign Languagerecognition system with two hand gestures.
7. Emphasize the significance of the system's accuracy and real-time performance in making sign language more accessible and inclusive.
8. Establish possible areas for improvement, like growing the dataset, improving the robustness of the system, or improving the algorithms.
9. To improve its usefulness for the Marathi Deaf community, Marathi Sign Language Recognition System can be integrated with other technologies like text-to-speech systems or communication devices.
10. Discuss the long-term sustainability of the system, including maintenance, updates, and ongoing support for users.

Chapter 8

Conclusion

8.1 Conclusion

The Marathi Sign Language Recognizer is like a smart tool that helps deaf people who use Marathi Sign Language communicate better. It can turn their signs into spoken words, making it easier for them to talk to people who don't understand sign language. It's like a bridge between two different ways of communicating. Additionally, it can also read written Marathi text out loud, which can be helpful for people who have trouble reading or seeing. So, it's not only for the deaf but also for others who may need assistance with reading. This technology is a big step towards making communication easier and more inclusive for everyone.

In conclusion, we accomplished an impressive accuracy for our Marathi Sign Language recognition system. Furthermore, we have shown that our system can process signs in real-time, calculating milliseconds per every frame. The system's features, which are and respectively, show how well it can identify Marathi Sign Language gestures while reducing false positives. The Marathi Deaf community's accessibility and communication could be greatly enhanced by these findings. There are obstacles to overcome, though, such as variances in sign language and possible cultural disparities. Ongoing initiatives should concentrate on dataset expansion, user education, and cross-cultural adaptation in order to optimize the system's impact. Through the integration of technology, a dedication to inclusivity, the Marathi Sign Language will be enhanced.

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